

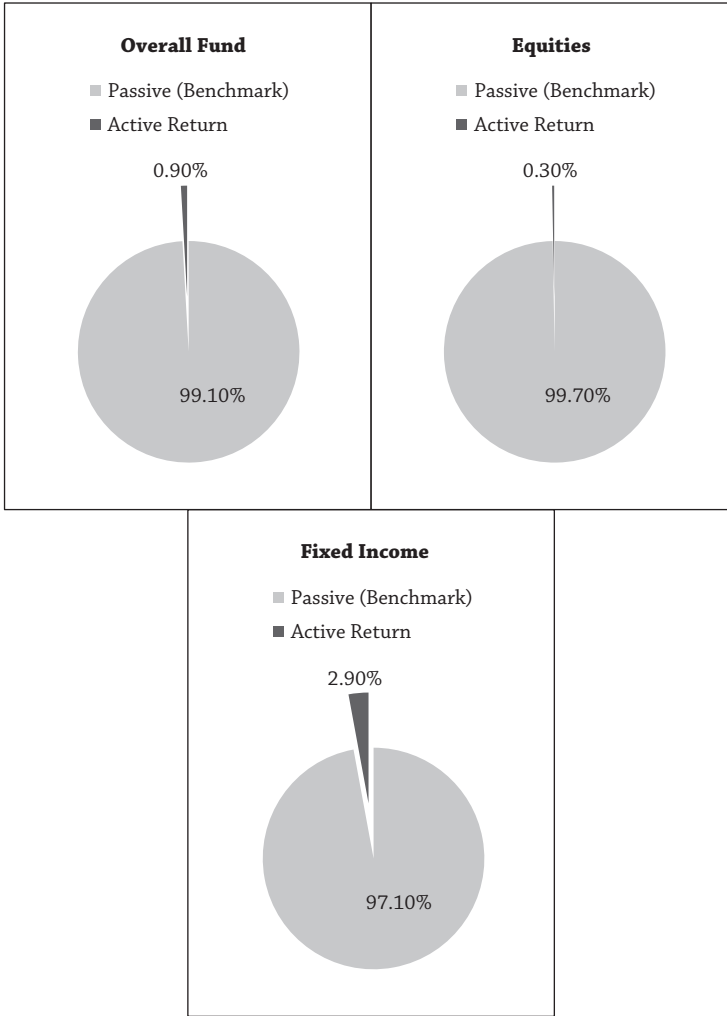


Figure 14.2

A decomposition of a total fund return into a benchmark return, a component where the asset classes in the benchmark are timed (called *tactical asset allocation*) and a component where securities are picked within each asset class (called *security selection*) is given by:

$$r_i = r_{bmk} + r_{timing} + r_{selection}. \quad (14.3)$$

In its full glory, this can be written as

$$r = \sum w_i r_i = \underbrace{\sum w_i^b r_i^b}_{r_{bmk}} + \underbrace{\sum (w_i - w_i^b) r_i^b}_{r_{timing}} + \underbrace{\sum w_i (r_i - r_i^b)}_{r_{selection}},$$